

ATP Creativity 02

Measuring the Creativity of an Automated Theorem Prover

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Abstract

This note converts the creativity controls introduced in *ATP Creativity 01* into quantities that can be estimated and optimized. It distinguishes a formal-system-relative scalar control index, which summarizes how exploratory a search policy is, from an empirical creativity score, which measures whether that policy produces verified and structurally novel proofs under a fixed budget. The construction is explicitly a data-science problem: raw search settings become engineered features, weights are calibrated on held-out theorem sets, repeated ATP trials estimate a noisy response surface, and an optimizer searches for configurations with high verified creative yield. Creativity need not be the sole objective of an ATP; success probability, computational cost, robustness, and proof quality may instead be combined through multi-objective optimization.

1. Control settings and observed creativity are different

Let the raw search-control vector be $\mathbf{u} = (\tau, w, b, n, r, \lambda, l)$, where the coordinates represent action-selection temperature, candidate width, breadth-versus-depth allocation, novelty bonus, restart or branch-abandonment rate, willingness to construct intermediate lemmas, and proof-length penalty.

Large control values do not imply successful creativity. Maximal temperature, width, or restarting can produce disorganized search and no proof. We therefore distinguish a control index $c_F(\mathbf{u})$, summarizing exploratory intensity in formal system F , from an empirical creativity score $C_{F,B}(\mathbf{u})$, measuring verified creative performance on benchmark B . The first is an input summary; the second is the quantity to optimize.

2. Formal-system-relative feature scaling

The raw coordinates have different units and cannot be added directly. For every coordinate define a calibrated transformation $z_{i,F}(\mathbf{u}_i)$ in $[0,1]$. Dependence on F is appropriate because the same numerical setting can induce different behavior in different proof calculi or theorem libraries.

$z_{\tau,F}$ = normalized action-selection entropy $z_{w,F} = \log(1+w) / \log(1+w_{\max},F)$ $z_{b,F}$
= fraction of budget allocated toward breadth $z_{n,F}$ = normalized ranking effect of
novelty bonus $z_{r,F}$ = normalized branch-abandonment frequency $z_{\lambda,F}$ = fraction of
effort devoted to lemma construction z_l,F = normalized tolerance for longer proof
routes

Behavioral normalizations are preferable to raw parameter normalizations whenever possible, because they are more comparable across implementations.

3. A calibrated scalar control index

$$c_F(\mathbf{u}) = \sum_i \alpha_{i,F} z_{i,F}(\mathbf{u}_i) \quad \alpha_{i,F} \geq 0, \quad \sum_i \alpha_{i,F} = 1$$

Then $c_F(\mathbf{u})$ lies in $[0,1]$. Equal weights provide a transparent baseline. Better weights can be estimated from controlled ablations or a predictive model trained to estimate verified novelty from normalized controls.

This average is deliberately *value-laden*: its transformations and weights encode which behaviors count as structured exploration. That is not automatically statistical bias. Leakage and overfitting arise when the same theorem outcomes are used both to choose the transformations or weights and to report final performance. Calibration, validation, and final evaluation must therefore use separate theorem sets. The full normalized vector should always be retained and reported.

4. Verified creative yield

For target theorem ϕ , information condition K_j , training exposure p , and fixed budget T , let $D_{F,\phi}(u)$ be the probability that a verified proof is found within T . Assign every normalized verified proof P a structural-novelty score $nu_F(P)$ in $[0,1]$, and let $N_{F,\phi}(u)$ be expected novelty conditional on success.

$$C_{F,\phi}(u) = D_{F,\phi}(u) N_{F,\phi}(u)$$

The score is near zero when unusual behavior rarely produces a proof and also when a reliable system merely reconstructs a familiar route. It rewards verified novelty rather than randomness or success alone.

5. Scarcity weighting and benchmark aggregation

Let $R_F(\Gamma, \phi)$ be the shortest-proof scarcity index from ATP Creativity 01. To keep one extreme or uncertain estimate from dominating a benchmark, use a bounded transform with $\rho_F > 0$:

$$R_{\text{tilde}}_F(\Gamma, \phi) = R_F(\Gamma, \phi) / (R_F(\Gamma, \phi) + \rho_F)$$

For benchmark B and predeclared theorem weights $q_\phi > 0$, define:

$$C_{F,B}(u) = [\sum_\phi q_\phi D_\phi(u) N_\phi(u) R_{\text{tilde}}_\phi] / [\sum_\phi q_\phi]$$

The separate components D , N , and R_{tilde} must also be reported, because systems can obtain the same product for different reasons.

6. The data-science formulation

ATP object	Data-science role
raw controls u	input variables
normalized controls z_F	engineered features
weights $\alpha_{i,F}$	calibrated parameters
repeated seeded runs	stochastic observations
verified proof outcome	binary response
structural novelty	conditional response
creative yield $D N$	objective variable
held-out theorem families	validation and test data

A rigorous workflow freezes the formal environment, partitions theorem families into calibration, validation, and test sets, samples control vectors, performs repeated seeded trials, verifies and normalizes all successful certificates, estimates uncertainty, fits a response surface, and evaluates the selected configuration once on the untouched test set.

Because ATP experiments are expensive and noisy, Bayesian optimization, Gaussian-process surrogates, tree-structured Parzen estimators, or other sample-efficient sequential methods are natural choices.

$$u^* \text{ in } \operatorname{argmax}_{u \text{ in } U_F} C_{F,B}(u)$$

There is no need to optimize c_F itself. It summarizes the selected controls; $C_{F,B}$ measures whether they work.

7. Creativity is not necessarily the best overall objective

A highly creative ATP may be unreliable, slow, unstable across seeds, or prone to producing unnecessarily complicated proofs. Let S be verified success rate, K normalized computational cost, V run-to-run instability, and Q proof quality or simplicity.

$$U_{F,B}(u) = \beta_C C_{F,B}(u) + \beta_S S(u) + \beta_Q Q(u) - \beta_K K(u) - \beta_V V(u)$$

A research prover may assign a large β_C ; a production prover may value reliability and cost more heavily. A safer alternative is to estimate a Pareto frontier rather than collapse every goal into one number. This exposes the tradeoff between

creativity and dependable theorem proving.

8. Recommended reporting standard

1. raw and normalized control vectors
2. the scalar control index, transformations, and weights
3. per-target and aggregate D, N, D N, and scarcity values
4. confidence intervals across independent seeds
5. calibration, validation, and test partitions
6. the optimizer and total tuning budget
7. all verified certificates and verifier/environment hashes
8. the complete distribution of proofs, bounded unknowns, and faults

9. Conclusion

The controls can be reduced to a formal-system-relative scalar, but that scalar measures exploratory settings rather than creativity itself. Measured ATP creativity is the production of verified structural novelty under frozen information and resource constraints. Its noisy response surface can be optimized with standard data-science methods. The strongest ATP, however, need not be the one with the highest creativity score; the appropriate objective depends on whether the intended use values discovery, reliability, speed, proof simplicity, or a declared combination of them.